Participant Workbook

Scientific Communication: Writing, Slide Design and Presentation Skills

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About this workshop

You already know your subject. This workshop helps you explain it clearly and keep your claims in line with your evidence. It is an introduction with guided practice over three hours, so it will not make anyone a finished writer or speaker. You will leave with four things you have made yourself: a revised paragraph, a short presentation storyboard, one redesigned slide and a rehearsed explanation.

Use your own work if you can, with any identifiers removed. If you cannot, the two fictional cases in this workbook will serve. Nothing in this workbook comes from a real study.

Learning outcomes

By the end of the workshop you should be able to:

- define your audience, purpose and main message.
- build an argument that connects a question, evidence, interpretation and implications.
- write clear scientific paragraphs, abstracts and slide headlines without overstating findings.
- design readable slides and explain figures accurately.
- deliver a timed presentation and respond to questions.
- use structured feedback to revise and rehearse your work.

Schedule

Time	Min	Block	What you will produce
0:00 to 0:08	8	Welcome and working pairs	-
0:08 to 0:28	20	Module 1: Audience, purpose and message	Audience-and-message worksheet
0:28 to 0:58	30	Module 2: Scientific writing	Revised paragraph
0:58 to 1:18	20	Module 3: Presentation structure	Short presentation storyboard
1:18 to 1:30	12	Break	-
1:30 to 2:04	34	Module 4: Slide design and data communication	One redesigned slide
2:04 to 2:14	10	Module 5: Delivery skills	Spoken opening line
2:14 to 2:36	22	Timed rehearsal in trios	Rehearsed explanation, peer feedback
2:36 to 2:48	12	Question-and-answer practice	Practised answers
2:48 to 2:55	7	Module 6: Accuracy and responsible use	Checked slide
2:55 to 3:00	5	Close: revise and commit	Personal revision plan
	180	**Total**	

How we treat evidence

This course separates three things. **Research findings** come from studies, and they have limits. **Professional advice** comes from experience, and most advice about slides is of this kind. **Our recommendations** are teaching devices that we find useful. You will hear each one labelled. Rules such as one slide per minute, six words per line or a minimum font size are rules of thumb. We did not find experiments that set those numbers, so test your slides in the real room.

The two fictional cases

Case A: CareCall

CareCall (fictional). Two hospitals looked at whether a nurse telephone call within 72 hours of discharge was linked to fewer unplanned readmissions within 30 days. Of 1,037 adults discharged from medical wards in one year, 187 were excluded (62 transferred, 41 on palliative pathways, 84 with no working phone number), leaving 850. Calls were not randomised: patients received one when a nurse was free and the patient could be reached.

Table: unplanned readmission within 30 days. Fictional teaching data. Not a real study.

Group	Readmitted	Patients	Risk (%)
Call	60	412	14.6
No call	94	438	21.5
Call, high discharge risk	32	120	26.7
No call, high discharge risk	59	198	29.8
Call, lower discharge risk	28	292	9.6
No call, lower discharge risk	35	240	14.6

The risk difference was 6.9 percentage points (95% CI 1.8 to 12.0 lower with a call). Within risk groups, combined by the Mantel-Haenszel method, it was 4.3 points lower (95% CI 9.4 lower to 0.8 higher). Emergency visits without admission were 40 of 412 (9.7%) and 47 of 438 (10.7%), a difference of 1.0 points lower (95% CI 5.1 lower to 3.0 higher).

Case B: CoolRoom

CoolRoom (fictional). A school district painted reflective coating on the roofs above 12 classrooms and compared mean afternoon temperature over one hot term with 12 classrooms under standard roofs. Head teachers volunteered the coated rooms, so allocation was not random.

Facts you may use: 12 coated and 12 standard classrooms. Mean afternoon temperature 29.4 °C and 30.6 °C. Difference 1.2 °C cooler in coated rooms (95% CI 2.5 °C cooler to 0.1 °C warmer). Head teachers volunteered the coated rooms. Exam results were not measured.

Module notes

Module 1: Audience, purpose and message

Decide who is in the room, what they know, what they need and what should change afterwards. Only then write your message. A topic names an area, and a message makes a claim someone could challenge. The same finding needs a different message for a conference, a thesis committee and a hospital management team, although the evidence and the uncertainty stay the same.

Room	Who they are	What they should do	Message
Conference audience	Peers in the field, mixed methods knowledge	Consider a new explanation and ask how it applies to their setting	In two hospitals, a nurse call was linked to 6.9 fewer readmissions per 100, but much of that gap reflected who could be reached.

Room	Who they are	What they should do	Message
Thesis committee	Examiners who know the field and will probe your choices	Judge that the work is yours, sound and a contribution	I show that reachability confounds observational evaluations of follow-up calls, and I defend the stratified analysis that reveals it.
Hospital management	Decision-makers with budgets, little time, limited methods detail	Decide what to do next	We recommend a six-month staggered trial on two wards before funding full roll-out, because the benefit could be 9.4 fewer to 0.8 more readmissions per 100.

Module 2: Scientific writing

A clear paragraph makes one point, gives evidence, interprets it and says what follows. Keep results apart from interpretation. Give counts with their denominators, report the size of a difference with its uncertainty, and choose verbs your design can support. A non-significant result is rarely proof of no effect: state what the interval allows. Avoid giving a relative figure alone, because relative figures tend to look larger and persuade more.

Within a sentence, readers tend to look to the start for context and to the end for what is new. That is reasoned advice from Gopen and Swan, and they do not offer it as a rule. Plain language means fitting words to your reader. Specialists can keep specialist terms where those terms carry meaning.

Worked example: a results paragraph

All data are fictional. The original paragraph is short and confident, and it overstates.

Before

> It is a well-known fact that post-discharge follow-up is of great importance. The implementation of nurse telephone calls in our hospitals led to a highly significant reduction in readmissions of 32% (p<0.05), which proves that the intervention works and should be rolled out everywhere. There was no effect on emergency visits.

After

> We examined whether a nurse telephone call within 72 hours of discharge was associated with fewer unplanned readmissions. Of 850 adults discharged from medical wards at two hospitals, 60 of 412 who received a call (14.6%) and 94 of 438 who did not (21.5%) were readmitted within 30 days, a difference of 6.9 percentage points (95% CI 1.8 to 12.0). Patients who received calls were less often at high risk at discharge (29% v 45%). Within risk strata the combined difference was 4.3 percentage points (95% CI 9.4 lower to 0.8 higher). Because calls were not randomised, these results do not show that the calls caused the difference. Emergency visits without admission were 9.7% and 10.7%, but the interval (5.1 lower to 3.0 higher per 100) is too wide to rule out a useful benefit or a small harm.

Table: what changed and why.

What changed	Why
Removed the opening claim	"Well-known" and "great importance" give the reader nothing to check.
Replaced the verb	"Led to" and "proves" claim cause. The calls were not randomised, so the question is now stated as an association.
Added counts and denominators	32% sounds large and hides the base. 60 of 412 and 94 of 438 let a reader check it.

What changed	Why
Reported the absolute difference and its interval	A risk difference with a 95% CI shows size and precision. "Highly significant" shows neither.
Exposed the confounder	High-risk patients were harder to reach. Readers need this to judge the crude result.
Moved the recommendation out	A results paragraph reports. The recommendation belongs in the conclusion, where it can be argued.
Rewrote "no effect"	A wide interval around a small difference means the data cannot tell. It does not show the calls had no effect.

One finding in four forms

The same result, rewritten for four uses. Words change between versions. The size, the population, the design and the uncertainty do not.

Form	Text
Paragraph	The 'after' paragraph above.
Abstract passage	Results: Of 850 patients, 60/412 (14.6%) with a call and 94/438 (21.5%) without were readmitted within 30 days (risk difference 6.9 percentage points lower, 95% CI 1.8 to 12.0 lower). After stratifying by discharge risk the difference was 4.3 points lower (95% CI 9.4 lower to 0.8 higher). Conclusions: Follow-up calls were associated with fewer readmissions, but much of the gap may reflect lower risk among patients reached. A randomised evaluation is needed.
Slide headline	Patients who got a call had fewer readmissions, but the gap narrowed after accounting for risk
Spoken explanation	For every hundred patients, those who got a call had about seven fewer readmissions. But the nurses reached healthier patients more easily. When we compare like with like, the gap is about four per hundred, and the data fit anything from nine fewer to one more. So this is a promising signal. It is not yet proof.

Sentence repairs

Eight common problems, with an explanation of each repair.

Problem	Before	After	Why
Vague	Readmissions were considerably lower in the call group.	Readmissions were 6.9 percentage points lower in the call group (14.6% v 21.5%).	Name the size.
Wordy	Due to the fact that a large number of patients could not be contacted, it was not possible to make calls to all of them.	Nurses could not call every patient, so 438 received no call.	Fewer words, same fact, and a real verb.
Jargon	Post-discharge telephonic engagement modalities were operationalised by nursing staff.	Nurses telephoned patients after discharge.	Keep technical terms when they carry meaning. These do not.
Result or interpretation	The calls worked, with 14.6% readmitted.	14.6% of patients who received a call were readmitted. (Result.) This may reflect the call, or that healthier patients were easier to reach. (Interpretation.)	Mark where the data end and your reading begins.

Problem	Before	After	Why
Denominator	95 readmissions at Hospital A against 59 at Hospital B shows that A performs worse.	Hospital A readmitted 95 of 540 patients (17.6%) and Hospital B 59 of 310 (19.0%).	Counts without denominators reverse the picture here.
Uncertainty	The calls reduce readmissions by 4.3 percentage points.	After stratifying by discharge risk, the estimate was 4.3 points lower (95% CI 9.4 lower to 0.8 higher).	A single number hides how much the data allow.
Causal language	Calls prevent readmissions.	Patients who received calls were less often readmitted. We cannot say from this design whether the calls caused this.	Match the verb to the design.
Non-significant	The coating had no effect on temperature (p>0.05).	Coated rooms were 1.2 °C cooler on average, but the 95% CI ran from 2.5 °C cooler to 0.1 °C warmer.	"Not significant" is not "no effect".

Module 3: Presentation structure

A talk is one argument: question, evidence, interpretation, implication. Open with the problem and the question. Show the methods your audience needs to judge the evidence: who, what was compared, what was measured and the main threat to validity. Lead with your main finding, and put a limit that changes its meaning next to it. End with what the evidence supports.

A conference talk spends most time on findings and meaning. An organisational briefing leads with the recommendation. A thesis defence examines your contribution and your methodological choices, so prepare to explain why you chose your methods over the alternatives, and build backup slides for expected challenges. Requirements vary a great deal: a UK viva may be private with no required talk, while a public defence in the United States or Sweden may last an hour or several. Read your own regulations first.

Module 4: Slide design and data communication

Give each slide a headline that states the claim, and use the body to show evidence. One experiment supports this bundle, and it has critics, so treat it as a strong default. Reduce text by moving detail to your speech and notes, while keeping the numbers needed to judge the claim. Test contrast and legibility in the real room. Do not let colour be the only signal: add labels, shapes or line styles.

Choose a chart from your question. Bars encode length, so they start at zero. Dots and lines encode position, so a narrower labelled range can be right. Every figure should tell the audience its axes, units, denominators, uncertainty and source. The file Before_After_Slides.pptx holds six redesigns with notes.

Module 5: Delivery, rehearsal and questions

Explain each figure in four moves: what it shows, where to look, what you see and why it matters. Speak from understanding, with brief notes for the route. Rehearse aloud against the real limit, and cut content before you speed up. Accent is not a fault. Being understood is the goal, and pace, pauses, signposts and stress on key words all help.

With questions, listen to the end, repeat or rephrase, answer briefly, then check. Find the valid part of a critical question. Ask what an unclear question means. Say when something is out of scope, and offer a route. If you do not know, say so.

Module 6: Accuracy and responsible use

When you simplify, keep the size, the population, the direction and the certainty. Credit sources and collaborators, and get permission before you acknowledge someone. Protect people in your data. A rare case on a named ward can identify a patient, small numbers can expose a person, and data with codes in place of names are usually pseudonymised and not anonymous. Find your organisation's rule.

AI tools can help you edit and rehearse. You remain responsible for every fact, reference and claim, so open every reference and check every number against your source. Do not paste confidential data into tools your organisation has not approved, and follow your journal's or institution's rules on disclosure.

Worksheets

Worksheet 1: Audience and message

Use your own talk if you can. Otherwise use CareCall or CoolRoom and choose one room: conference, thesis committee, or hospital or organisation.

Occasion and time limit	
Who is in the room? (roles, field, seniority, language)	
What do they already know?	
What do they need from you?	
What should they understand, decide or do afterwards?	

Your message in one sentence

Check your sentence. Tick each box.

- It is a full sentence with a verb.
- Someone could disagree with it.
- You could say it in a corridor, without slides.
- It is no stronger than your evidence.

Pair check. Read your sentence aloud. Your partner says what they think you want them to do. What did they say?

Worksheet 2: Paragraph editing

Edit the paragraph below, or a paragraph of your own. Use only the facts given. Do not invent numbers. This is fictional teaching data.

> In light of the fact that excessive heat in classrooms constitutes a significant problem for learning, the implementation of a reflective roof coating was undertaken in a number of classrooms. It was found that temperatures were much lower in the coated rooms, showing that the coating cools classrooms and will improve exam results. However, the difference was not statistically significant ($p>0.05$), so it can be concluded that the coating does not work.

Facts you may use: 12 coated and 12 standard classrooms. Mean afternoon temperature 29.4 °C and 30.6 °C. Difference 1.2 °C cooler in coated rooms (95% CI 2.5 °C cooler to 0.1 °C warmer). Head teachers volunteered the coated rooms. Exam results were not measured.

Step 1. Underline at least five problems. Look for wordy phrases, vague claims, missing denominators, causal language, and anything stated but not measured.

Step 2. Rewrite it. Aim for one point, then evidence, then interpretation.

Step 3. Swap with your partner. Check the rewrite for three things: is the size given, is the uncertainty given, and is any claim stronger than the data?

Worksheet 3: Storyboard

Write one full-sentence headline for each card, then sketch the evidence under it. Six cards is a device for today, not a rule for your talk. Do not design slides yet.

1 Problem and question	2 What we did
3 Main finding	4 Limit or second finding
5 What it means	6 Conclusion or recommendation

Title test. Read your headlines aloud in order. Do they tell the story without slides? Which card would you cut if you had half the time?

Worksheet 4: Slide critique and redesign

The slide is on screen, and it is also in Slide_Redesign_Exercise.pptx. Critique it in pairs for 3 minutes, then redesign it alone for 9 minutes. All values are fictional.

Discharge risk	Call	No call
High	32 of 120 (26.7%)	59 of 198 (29.8%)
Lower	28 of 292 (9.6%)	35 of 240 (14.6%)

What is the slide's claim?	
What evidence supports it?	
What would a viewer with a colour vision difference lose?	
What does 'NS' hide?	

Your headline, as a full sentence

Sketch your evidence. Keep the denominators and the uncertainty.

Worksheet 5: Rehearsal checklist and feedback

Before you rehearse

- State your main message in one sentence.
- Mark each slide: must keep, should keep, can cut.
- Practise your opening and your last sentence aloud.
- Practise any technical term or long number you find hard to say.
- Write the time you should reach at each section.

The six-minute round in trios

Minutes	Speaker	Listeners
0:00 to 2:00	Explain your slide using the four moves: what it shows, where to look, what you see, why it matters.	Listen. The timekeeper stops the speaker at 2:00.
2:00 to 3:00	Write down what you would change.	Complete the prompts below in silence.
3:00 to 5:00	Listen. Ask for clarification only.	Give feedback using the prompts.
5:00 to 6:00	Repeat your weakest part once.	Say whether it was clearer.

Feedback prompts, for each speaker

The main message I understood was:	
I stopped following when:	
The one change that would help most is:	

Keep feedback on clarity and understanding. Do not comment on accent, looks or personality.

Worksheet 6: Feedback rubric

Use this rubric to give feedback on practice. It is a course feedback tool, not a validated assessment instrument, and it should not be used to grade people or to make decisions about them. It rates the work and never accent or personality. Choose the description that fits best and give one example from what you saw or heard.

Criterion	1 Not yet	2 Developing	3 Secure	Example
Message	No single claim can be identified, or the listener names a different one from the speaker.	A claim is present, but it drifts, or the listener needs help to repeat it.	One clear claim. The listener can repeat it, and it suits this audience and purpose.	
Structure	Reads as a list of topics. Steps do not follow from each other.	Most steps link. One gap between question, evidence, interpretation or implication.	Question, evidence, interpretation and implication connect, with clear transitions.	
Writing	Vague, wordy or overstated. Results and interpretation are mixed.	Clear in most places. Some vague wording, or one claim slightly too strong.	Clear and direct. Each sentence is no stronger than its evidence.	
Evidence	Numbers without context, or a claim with no numbers. Non-significance read as no effect.	Most of size, denominator and uncertainty are given. Design only partly stated.	Size, denominator and uncertainty are given where relevant, with honest limits.	
Visual clarity	Hard to read from the back. Message not visible. Colour used as the only signal.	Readable, but cluttered, or the headline is a topic. Axes or units partly explained.	Readable, one claim visible, figure explained, and nothing relies on colour alone.	
Timing	Over the limit by more than a quarter, or stopped before the main point.	Within a quarter of the limit, but the ending was rushed or reached only just.	Within the limit, with the main point and conclusion delivered at an even pace.	
Responses to questions	Avoids, bluffs or argues. Does not state uncertainty.	Answers part of the question, or answers at length. Limits partly stated.	Listens, checks, answers briefly, states limits and says when something is unknown.	

What the ratings mean. 1 Not yet: the listener would struggle, or the claim is unsafe. 2 Developing: the main idea reaches the listener, with one clear thing to improve. 3 Secure: it works for this audience and could be used as it is. A rating of 3 does not mean finished.

Worksheet 7: Question cards

In trios, give your main message in one sentence. A partner reads a card, or writes a fair question of the same type. Answer in one minute: listen, repeat, answer briefly, check. Questions refer to CareCall; adapt them to your own topic.

Type	Question
Critical	Your groups were not comparable, so why should we believe any of this?
Critical	Isn't this just regression to the mean, or a healthier group being reached?
Critical	You have one year of data from two sites. Why generalise at all?
Unclear	What about the, you know, the other thing with the numbers?
Unclear	Can you say more about the difference?
Unclear	Doesn't that depend on the setting?
Out of scope	How would this work in primary care?
Out of scope	What is the cost per readmission avoided?
Out of scope	What do you think of the new national funding policy?
Unknown	What was the result for patients aged over 85?
Unknown	Has anyone tested this in a low-income setting?
Unknown	Which statistical package did your colleague use for that sensitivity analysis?

Phrases you can use

Moment	Phrase
To gain time	"Let me check I understood: you are asking about..."
To clarify	"Do you mean the crude result or the stratified one?"
When criticised	"That is a fair concern. Here is what we did about it, and what remains uncertain."
Out of scope	"That is outside this study. I can follow up afterwards."
When you do not know	"I do not know. The data cannot answer that. I will check and reply."
To state uncertainty	"Our best estimate is X, and the data fit anything from Y to Z."

Personal plan and course feedback

My plan

The one thing I will revise first	
My rehearsal: with whom, and on what date?	
The claim I will verify	

Accuracy check for any slide

- Every number matches the source table.
- Denominators and units are shown.
- Uncertainty is shown.
- The verb fits the design.
- The headline is no stronger than the data.
- No person can be identified.
- Sources and collaborators are credited.
- AI use is disclosed, if required.

Course feedback

Please tick one for each statement	Disagree	Neutral	Agree
The session was useful for my work.			
The exercises fitted the time.			
Feedback focused on clarity and understanding.			
I would recommend this workshop.			

What should we change?

Where the guidance comes from

The full reference list, with links, is in References_and_Source_Map. Useful starting points are Bourne (2007), Naegle (2021), the assertion-evidence site at assertion-evidence.com, the CDC Clear Communication Index, and Altman and Bland (1995).